

Applications of Poincaré Boundary Condition for 3D Ideal MHD Equilibrium and Optimization

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Stellarator ideal MHD codes such as VMEC and DESC solve the equilibrium problem by prescribing the last closed flux surface (LCFS), a choice that determines the field uniquely only in vacuum and that guarantees neither the existence of nested flux surfaces nor a unique equilibrium at finite β . In this paper, we propose prescribing the Poincaré cross-section at a single toroidal plane instead to break the unnecessary dependence on LCFS, and implement it in DESC. Constraining the geometry on one plane rather than on a full toroidal surface leaves more spectral coefficients free, and we show examples of solving an equilibrium with its $\zeta = 0$ cross-section held fixed, starting either from scratch or from an existing LCFS solution. As a standalone solver the method is more exposed to the local minima of the force-error minimization, and therefore more dependent on the initial guess, but as a refinement step applied to an existing LCFS solution it reliably lowers the volume-averaged normalized force error by an order of magnitude. We further show that the Poincaré coefficients can be used directly as design variables by optimizing a quasi-helical configuration that maintains high-fidelity force balance throughout the process.

Key words: MHD Equilibrium, Stellarators, Numerical Methods, Optimization

1. Introduction

Stellarators are a class of magnetic confinement fusion devices that use complex, three-dimensional magnetic fields to confine plasma. Unlike tokamaks, they do not require a large internal plasma current for confinement, which grants them inherent stability against major current-driven disruptions (Helander 2014). The intricate 3D nature of the stellarator magnetic field makes analytical solutions to the governing ideal magnetohydrodynamic (MHD) equations intractable (Freidberg 2014), except in simplified regimes like near-axis expansions or high aspect ratio. Consequently, progress in the stellarator field has been driven primarily by numerical simulations.

The vast design space of stellarators presents promising candidates for a viable fusion reactor, making optimization methods essential. Equilibrium codes like VMEC (Hirshman & Whitson 1983) and DESC (Dudt & Kolemen 2020; Dudt *et al.* 2023; Panici *et al.* 2023; Conlin *et al.* 2023) are frequently used in these optimization studies because the assumption of nested magnetic flux surfaces makes the computation more efficient. While higher-fidelity codes that do not assume nested surfaces exist (e.g., SPEC (Hudson *et al.*

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2012)) and have been incorporated in the optimization studies (Landreman *et al.* 2021; Baillod *et al.* 2022), their computational cost currently limits their use in large-scale optimization.

Equilibrium solvers typically leverage the toroidal periodicity of the system to represent flux surfaces spectrally with a double Fourier series. The equilibrium solution is then found by optimizing the spectral coefficients to minimize either the total MHD energy or the ideal MHD force error, $\|\mathbf{J} \times \mathbf{B} - \nabla p\|$. This calculation requires a set of inputs: a pressure profile, a current or rotational transform profile, the total enclosed magnetic flux, and the shape of the last closed flux surface (LCFS), the toroidal boundary to which the magnetic field is tangent (Kruskal & Kulsrud 1958). The assumption underlying this construction, that the volume is covered everywhere by nested flux surfaces, is a strong one, and it is not satisfied by a general three-dimensional field: islands and stochastic regions are generic features, and their absence for a given LCFS is not guaranteed (Grad 1967; Garabedian 2003). Solvers of this class therefore do not guarantee an island-free field. They return the best nested-surface representation available at the given resolution, and the residual force error that they report is in part a measure of the field structure that such a representation cannot express (Lazerson *et al.* 2016).

Once the solution is understood as an approximation to a field that is not globally nested, prescribing the LCFS becomes one of several ways to close the problem rather than a privileged one. Its strongest justification is in vacuum, where the field satisfies Laplace’s equation and the boundary shape, together with the total enclosed magnetic flux, determines the solution uniquely. At finite β , which is the regime that stellarator optimization ultimately targets, no such guarantee exists. Distinct equilibria can share the same LCFS, as in the tokamak states with an axisymmetric boundary and a three-dimensional helical core (Cooper *et al.* 2010; Panici *et al.* 2026), where the magnetic axis acquires a helical excursion that the boundary shape does not reveal; other bifurcated equilibria have also been discussed in the literature (Garabedian 2003; Ilgisonis & Pozdnyakov 2004; Garabedian 2006). The analytical appeal of the LCFS is therefore confined to a limit that optimization studies are moving away from.

In this paper we propose a novel alternative: prescribing the Poincaré cross-section of the field at a single toroidal plane (e.g., $\zeta = 0$), which we refer to as the Poincaré boundary condition. Prescribing the geometry on a single plane rather than on a full toroidal surface leaves more of the spectral coefficients free, giving both the equilibrium solver and an optimizer built on it a larger space to search. Because DESC represents the equilibrium spectrally in all three dimensions, this condition enters the solve only through its linear constraints and therefore costs no more than a fixed-LCFS solve, whereas it would not be as straightforward in codes that discretize the radial direction on a grid of flux surfaces. This is an algorithmic choice rather than an analytical one, and we return in the next section to what it does and does not imply.

This paper is organized as follows. First, we will detail the numerical implementation of this new boundary condition within the DESC code. Next, we will demonstrate its versatility by exploring several use cases: solving for MHD equilibrium, refining existing equilibria to reduce force errors, and finally, incorporating the Poincaré section shape as a variable in optimization studies.

2. Numerical Model

DESC is a fully spectral code and represents the shape of the nested flux surfaces, R and Z , as well as the poloidal stream function λ by a Fourier-Zernike basis in the full toroidal domain. The map from (ρ, θ, ζ) , where θ is an arbitrary poloidal angle and ζ is

equal to cylindrical coordinate ϕ , to a general function f (which is either R, Z or λ) can be written as,

$$f(\rho, \theta, \zeta) = \sum_l \sum_m \sum_n f_{lmn} \mathcal{R}_l^{|m|}(\rho) \mathcal{F}^m(\theta) \mathcal{F}^n(\zeta) \quad (2.1)$$

where f_{lmn} are the spectral coefficients, $\mathcal{F}^m(\theta)$ and $\mathcal{F}^n(\zeta)$ are the Fourier components corresponding to poloidal and toroidal mode numbers, respectively, and $\mathcal{R}_l^m$ is the radial part of the Zernike polynomials which is defined as,

$$\mathcal{R}_l^{|m|}(\rho) = \sum_{s=0}^{(l-|m|)/2} (-1)^s \binom{l-s}{s} \binom{l-2s}{\frac{l-|m|}{2}-s} \rho^{l-2s} \quad (2.2)$$

where the mode numbers l and m are defined such that $l \geq 0$, $|m| \leq l$, and the difference $l - |m|$ must be even (Zernike 1934; Zernike & Stratton 1934). For the LCFS where $\rho = 1$, the radial part of the Zernike polynomials is always equal to 1. This results in a simple linear relation between the LCFS boundary and global coefficients,

$$Rb_{mn} = \sum_l R_{lmn} \quad Zb_{mn} = \sum_l Z_{lmn} \quad (2.3)$$

where Rb_{mn} and Zb_{mn} are the LCFS boundary coefficients, which are either given by the user or optimized for some objective.

The way equilibrium is solved in DESC can be summarized by the following optimization problem,

$$\begin{aligned} \min_x & \|\mathbf{J}(x) \times \mathbf{B}(x) - \nabla p(x)\|_2^2 \\ \text{s.t. } & \mathbf{A}x = b \end{aligned} \quad (2.4)$$

where $\mathbf{A}$ is the linear constraint matrix composed of Equation 2.3 along with other constraints (such as fixed profiles, gauge freedom of λ etc.) and x is the state vector defining the equilibrium. The linear constraint can then be eliminated by writing $x = \mathbf{N}y + x_p$, where $\mathbf{N}$ is a basis for the nullspace of $\mathbf{A}$ so that $\mathbf{A}\mathbf{N} = 0$, y is the reduced state vector, and x_p is a particular solution of $\mathbf{A}x_p = b$.

To adapt the equilibrium solution to the Poincaré boundary condition, we expand the prescribed cross-section in the same basis restricted to that plane, with coefficients Rp_{lm} , and evaluate Equation 2.1 at $\zeta = 0$,

$$\sum_l \sum_m Rp_{lm} \mathcal{R}_l^{|m|}(\rho) \mathcal{F}^m(\theta) = \sum_l \sum_m \left(\sum_n \mathcal{F}^n(\zeta = 0) R_{lmn} \right) \mathcal{R}_l^{|m|}(\rho) \mathcal{F}^m(\theta) \quad (2.5)$$

Since the Zernike polynomials and the Fourier series are linearly independent, we have,

$$Rp_{lm} = \sum_n R_{lmn} \mathcal{F}^n(\zeta = 0) \quad (2.6)$$

and,

$$\mathcal{F}^n(\zeta = 0) = \begin{cases} 0 & \text{for } n < 0 \\ 1 & \text{for } n \geq 0 \end{cases} \quad (2.7)$$

The same relations hold for Z and for the poloidal stream function λ , and the constraint on λ can be imposed or omitted independently of the geometric one. Therefore, the

implementation of the new boundary condition only changes the linear constraint matrix $\mathbf{A}$ in Equation 2.4, hence most of the machinery in DESC can be kept unchanged.

It is worth noting that, due to the relation in Equation 2.7, evaluating at $\zeta = 0$ removes the sin modes from the constraint entirely and collapses the cos modes into a single sum for each (l, m) . The Poincaré boundary condition therefore fixes one linear combination of the toroidal coefficients belonging to a given (l, m) and leaves the rest free. The constraint matrix consequently has fewer rows than its LCFS counterpart and a correspondingly larger nullspace[†], and hence more free parameters. Although we do not share results of it in this work, one can prescribe a second cross-section at another toroidal plane, such as $\zeta = \pi/(2N_{fp})$, which brings some of the sin modes into the constraint as well.

This counting describes the space in which the solver searches, and not the set of equilibria contained in it. Force balance must still be satisfied, and it is what selects a solution within that space, so a larger nullspace does not by itself imply that more than one equilibrium shares a given cross-section. What the extra freedom does change is the landscape of the minimization. Minimizing the force error is a non-convex problem with multiple local minima, and the more free directions the solver is given, the more of them it can reach. These minima are not peculiar to the Poincaré boundary condition; they are present under a fixed LCFS as well. There, however, the smaller search space and the customary initial guess, obtained by scaling the boundary surface inward about an assumed axis, keep the solver in the vicinity of a single one of them. Under the Poincaré boundary condition the same minima become reachable, and different initial guesses can converge to different low force-error states that share the prescribed cross-section.

These minima can differ substantially in character. Two solutions that satisfy the same prescribed cross-section may have very different LCFS shapes, volumes and residual force errors, and more of them are within reach than with a prescribed LCFS, since the Poincaré boundary condition is the less restrictive of the two and constrains the geometry on a single plane rather than on a full toroidal surface. We treat this as a means of exploring the design space rather than as a defect: the additional free coefficients are what allow the solver to reach a lower $\|\mathbf{J} \times \mathbf{B} - \nabla p\|$, and what allow an optimizer to reach configurations that a rigid LCFS would prohibit.

As stated in the introduction, this choice is algorithmic rather than analytical. Neither boundary condition guarantees an island-free field, and fixing a cross-section says nothing about whether nested surfaces exist. In either case the solver returns the best nested representation available at the given resolution. If relaxing the constraint lowers the force error, one might ask why we keep a boundary condition at all, since removing it entirely and leaving every spectral coefficient free would lower the force error further. The reason is that the boundary condition is also what tells the solver which device we are looking for. The prescribed cross-section still sets the size and shape of the plasma in that plane, such as its minor radius and elongation, while the geometry away from the plane is left free.

[†] The exact dimensions depend on the equilibrium resolutions L , M and N . For the stellarator-symmetric case with $L = M = 12$ and $N = 6$, the resolution used for the quasi-helical optimization of §3.3, the Poincaré boundary condition imposes 49 constraints on the 595 spectral coefficients of R , whereas the LCFS imposes 163.

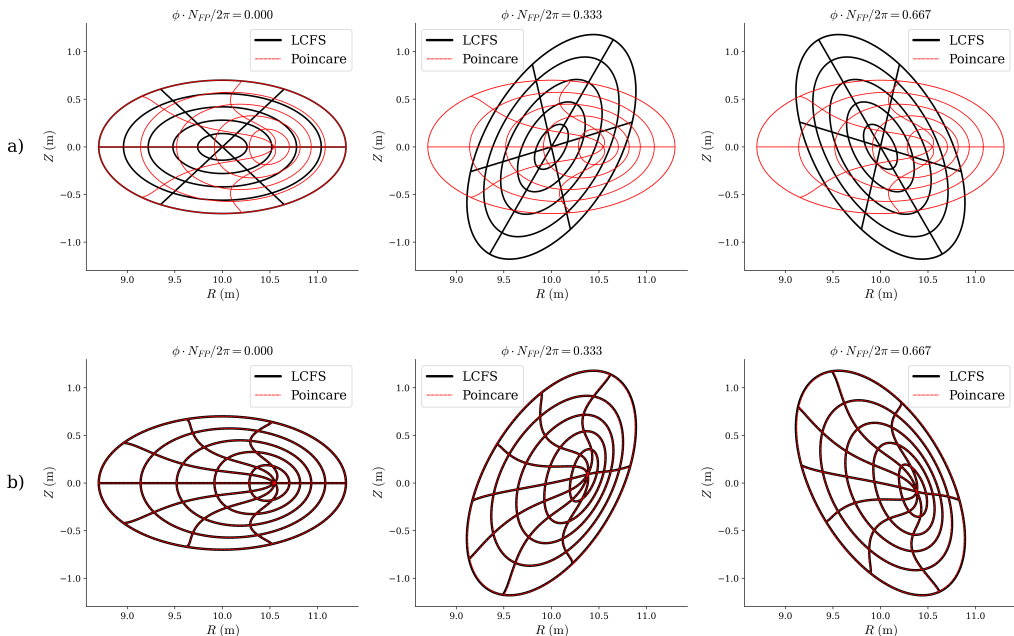


Figure 1: The initial (top) and final (bottom) flux surface shapes for LCFS (red) and Poincaré boundary conditions (black) of the HELIOTRON device with $N_{fp} = 19$. The lines that cross the constant ρ surfaces are the constant straight field line angle $\vartheta = \theta + \lambda$.

3. Results

3.1. Equilibrium Solve

As described in the previous section, one of the inputs to an equilibrium solve is the boundary shape. When the boundary is the LCFS, Equation 2.3 defines some of the R and Z coefficients, and the rest can be initialized by scaling down that surface around an initial axis, which for convenience is taken to lie midway between the $\theta = 0$ and $\theta = \pi$ curves of the boundary. For the poloidal stream function λ , there is no good a priori guess, so it is usually set to 0. On the other hand, for the Poincaré boundary condition, only the cross-section at $\zeta = 0$ is known, which here is taken from the $\zeta = 0$ cut of the original equilibrium, so the natural initial guess is to revolve it toroidally and start from an axisymmetric shape with $N = 0$. The poloidal stream function λ can be fixed or left free at the cross-section[‡]. The initial guess surfaces for the HELIOTRON device with $N_{fp} = 19$ and $\beta = 10\%$, solved at a resolution of $L = 24$, $M = 12$ and $N = 3$, are shown in Figure 1 for both LCFS and Poincaré boundary conditions with fixed λ .

The equilibrium solve is conducted in stages, increasing the toroidal resolution N before each step until the resolution of the original equilibrium is reached. In Figure 1, it can be seen that the final surface shapes are visually similar. Moreover, the obtained equilibrium has good force balance. In Figure 2, we observe that in the LCFS solution, the error is accumulated around the LCFS boundary, whereas in Poincaré solution, this trend is observed more around the fixed cross-section. Final volume averaged normalized force error is $3\text{E-}4$ and $1.7\text{E-}4$ for LCFS and Poincaré, respectively.

Although the final equilibria are very similar for the HELIOTRON device with $N_{fp} =$

[‡] When λ is free, only the geometry of the flux surface shapes is given, whereas the inclusion of λ also provides information on the straight field line angle ϑ .

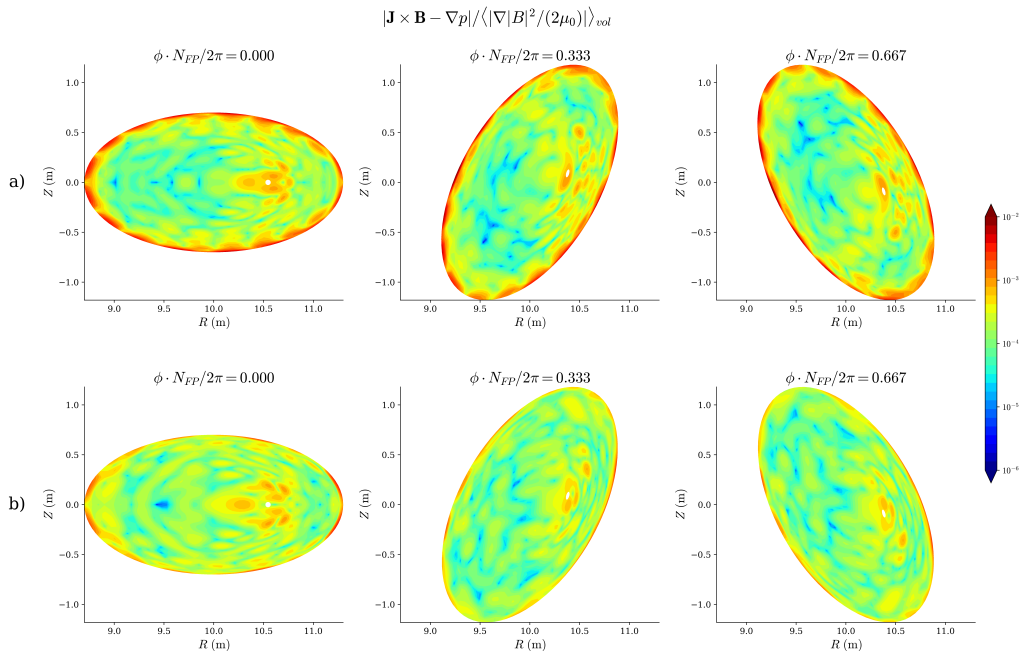


Figure 2: The ideal MHD force error normalized by volume-averaged magnetic plasma pressure for (top) the Poincaré solution and (bottom) the LCFS solution of the HELIOTRON device with $N_{fp} = 19$. The errors are plotted for 3 equispaced toroidal angles between 0 and $2\pi/N_{fp}$ in log scale, over the range 10^{-2} to 10^{-6} .

19, we see that as the LCFS boundary has more shaping, i.e., higher Fourier harmonics for toroidal terms, the axisymmetric initial guess tends to give different equilibria, and in many cases the final equilibrium expands well beyond the original device. Part of this behaviour originates in the objective function rather than in the boundary condition itself. With the LCFS prescribed, the enclosed cross-sectional area is fixed, and at fixed toroidal flux Ψ so is the field strength, since $B \sim \Psi/A$; expansion is simply not available to the solver, and the force error can therefore be normalized once at the start of the solve, as is done in DESC. Under the Poincaré boundary condition the equilibrium is free to expand away from the prescribed plane, resulting in a weaker field, and because $\|\mathbf{J} \times \mathbf{B}\|$ scales as B^2 , an inflated equilibrium computes a smaller residual against a normalization that no longer reflects its field strength. Normalizing by a quantity recomputed at each iterate, such as the volume-averaged B^2 , can help here, since a measure of that kind is unchanged by a uniform weakening of the field. For the same reason, every force error quoted in this paper is normalized by the volume-averaged magnetic pressure of the equilibrium it refers to, so that the comparisons that follow are not affected by differences in size or field strength between solutions. The volume and aspect ratio that a prescribed LCFS carries implicitly can likewise be supplied explicitly as additional objectives.

The gap in the number of free parameters also widens as N grows relative to L , since the LCFS constrains one coefficient for every (m, n) pair whereas the cross-section constrains one for every (l, m) pair, and a high toroidal resolution is usually needed for low N_{fp} devices. Combined with an initial guess that is far from the intended configuration, this leads the solver to a different local minimum, though not necessarily to a worse one, since the additional freedom generally yields better force balance with the trade-off being a

very different LCFS shape. In the following section we show that beginning from an existing LCFS solution keeps the solution nearby while still improving the force balance.

3.2. Reduced MHD Force Error

In the previous section, we noted that the lack of a good initial guess for the Poincaré boundary condition usually results in a different final equilibrium. However, one can expect to overcome this by using the LCFS solution as the initial guess. The aim of this approach is to improve the relatively high force error for the fixed-LCFS solution observed in Figure 2 and Figure 3 around the LCFS. Since the Poincaré and LCFS boundary conditions act on different R_{lmn} and Z_{lmn} coefficients, this helps the optimizer to discover the previously forbidden regions of the optimization space while still being close to the LCFS solution. We therefore restrict the problem to a single question: whether an equilibrium that stays close to that solution, but has a lower force error, can be obtained from the Poincaré boundary condition alone, with no additional objectives imposed. Establishing this is a prerequisite for the optimization of §3.3, where the same freedom must keep the force error low while other design metrics are being targeted. For this section and later ones, we won't fix the poloidal stream function λ , in other words, we are just imposing a geometric constraint on flux surface shapes at $\zeta = 0$ toroidal cut without giving information on the straight field-line angles.

In Figure 3, we take the W7-X equilibrium at $\beta = 2\%$ and a resolution of $L = M = N = 12$, which is solved until the relative decrease in the objective value between two consecutive iterations is less than $1\text{E-}3$ and solve for the force balance while keeping the $\zeta = 0$ cross-section of the previous solution. The termination condition for the Poincaré solve is the same as the LCFS one. It can be seen that solving the equilibrium by imposing Poincaré boundary conditions improves the force balance substantially. The LCFS moves by 1.9% of the minor radius on average and by at most 7.4%, and the plasma volume changes by 0.6%. Only the cross-section that is fixed sees minor improvement, which is expected due to the limited freedom. As an optional step, we then feed this solution back to a fixed LCFS solver to smooth the force error. Depending on the equilibrium resolution, the final LCFS solve can achieve a uniform force error distribution throughout the volume. This procedure can be repeated multiple times to achieve better force balance without any other optimization metric. The volume-averaged force error normalized by volume-averaged magnetic plasma pressure for each step of the above procedure is $2.86\text{E-}3$, $2.70\text{E-}4$ and $2.09\text{E-}4$, respectively. Almost all of the improvement comes from the Poincaré step, with the subsequent fixed-LCFS solve contributing comparatively little. Since that last solve holds the new boundary while leaving the cross-section free, the Poincaré solution is, at the tolerance used here, already close to a minimum of the fixed-LCFS problem for its own boundary.

It is worth noting that for Figure 3, we didn't fix the λ coefficients at the $\zeta = 0$ cross-section. We observe that while imposing a hard constraint on λ still improves the force balance, the resulting improvement is less pronounced.

Admittedly, this procedure modifies the equilibrium, which may have been previously optimized for specific physics metrics; this shift potentially degrades those optimized values. Nevertheless, the approach remains valuable as it demonstrates that solving the equilibrium with Poincaré boundary conditions generally yields a superior force balance, which many physics metrics implicitly assume to hold exactly. In the following section, we will demonstrate how Poincaré variables can be integrated directly into the optimization process to maintain high-fidelity force balance throughout the entire procedure.

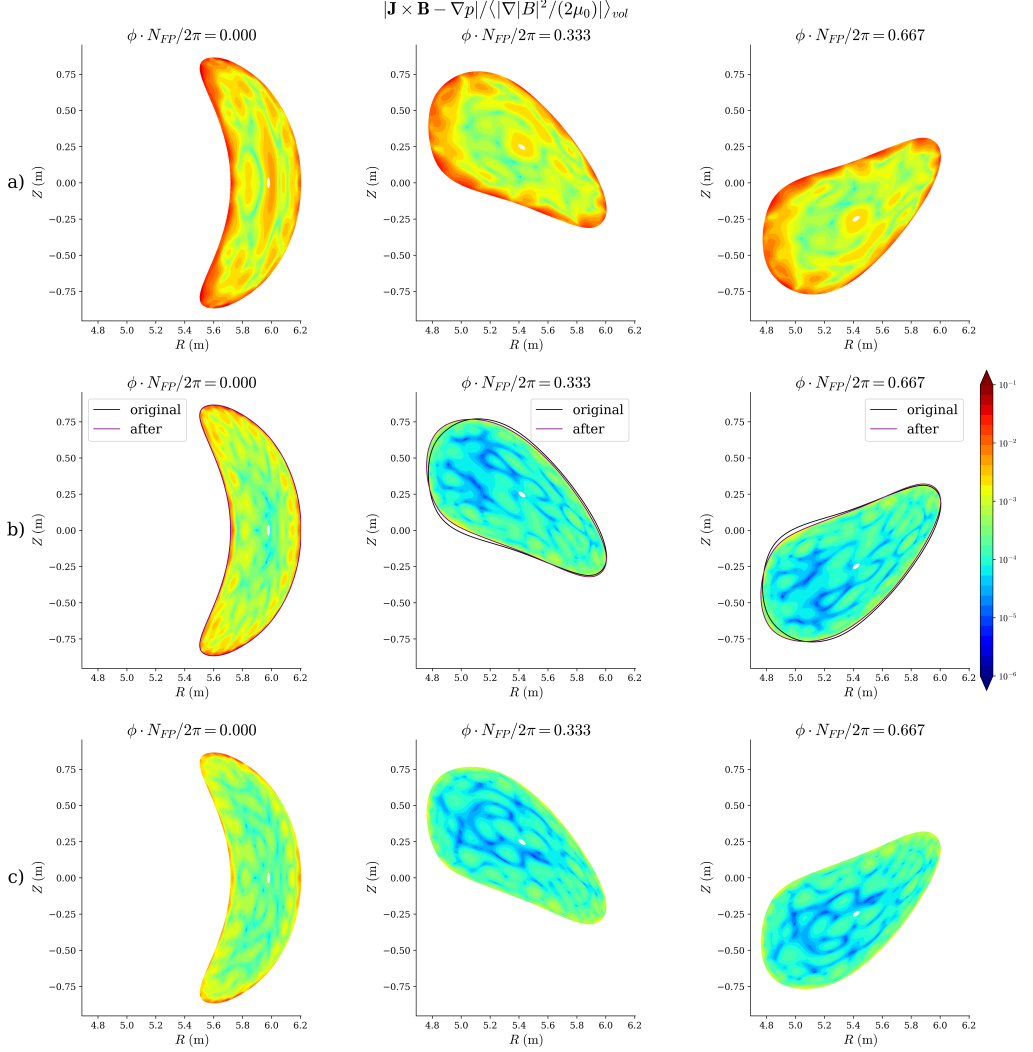


Figure 3: The ideal MHD force error normalized by volume-averaged magnetic plasma pressure (a) for LCFS solution, (b) Poincaré solution, (c) Poincaré solution fed back to LCFS solver of the W7-X device. All of them are solved for $\text{ftol}=1\text{E-}3$, which is the relative decrease in the objective value between 2 consecutive iterations. The errors are plotted for 3 equispaced toroidal angles between 0 and $2\pi/N_{fp}$. Part (b) additionally shows the LCFS of both equilibria for comparison.

3.3. Stage-1 Optimization

This section will present how the equilibrium optimization is conducted in DESC and how we modified it to use Poincaré variables. A general stage-1 optimization problem can be written as follows:

$$\begin{aligned}
 \min_{\mathbf{c}} \quad & f(\mathbf{x}, \mathbf{c}) \\
 \text{s.t.} \quad & g(\mathbf{x}, \mathbf{c}) = 0 \\
 & \mathbf{A}\mathbf{c} = \mathbf{b}
 \end{aligned} \tag{3.1}$$

where x includes the 3D spectral coefficients of the equilibrium, c is the optimizable parameters of the boundary from which we solve the equilibrium, f is the objective to minimize, g is the nonlinear ideal MHD equilibrium constraint and $\mathbf{A}$ is the linear constraint matrix. To deal with the non-linear constraint, DESC solves the equilibrium with the current c values to find the projected x values on to this non-linear constraint. Once the projected state is obtained, a gradient-based optimization can be conducted. We first expand the f and g using a first order Taylor series,

$$f(x + \Delta x, c + \Delta c) = f(x, c) + \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial c} \Delta c \quad (3.2)$$

$$g(x + \Delta x, c + \Delta c) = g(x, c) + \frac{\partial g}{\partial x} \Delta x + \frac{\partial g}{\partial c} \Delta c \quad (3.3)$$

By solving the equilibrium before and after, we aim for $g(x + \Delta x, c + \Delta c) = g(x, c) \sim 0$, which holds better for Poincaré solve. This results in the following relation for each optimization step,

$$f(x + \Delta x, c + \Delta c) = f(x, c) + \left[\frac{\partial f}{\partial c} - \frac{\partial f}{\partial x} \left(\frac{\partial g}{\partial x} \right)^{-1} \frac{\partial g}{\partial c} \right] \Delta c \quad (3.4)$$

The term in the square brackets is the Jacobian of the objective with respect to the optimization variables. Each optimization step tries to minimize $f(x + \Delta x, c + \Delta c)$, and this can be done in many different ways. The algorithm we use for the remaining of the optimizations is a trust-region based solver. To make this procedure work for the Poincaré variables, we have to update all the partial derivatives as well as replace the internal equilibrium solver to a fixed Poincaré section variant we discussed above.

Now, we will show an example optimization for quasihelical symmetry (QH). The exact problem is as follows,

$$\begin{aligned} \min_{c=(Rp_{lm}, Zp_{lm})} \quad & w_{qs} f_{qs}^2 + w_{ar} f_{ar}^2 + w_B f_B^2 \\ \text{s.t.} \quad & g(x, c) = 0 \quad Ac = b \end{aligned} \quad (3.5)$$

where $c = (Rp_{lm}, Zp_{lm})$ are the coefficients defining the Poincaré cross-section. The problem setup is very similar to the one employed in Landreman & Paul (2022). The objective function terms include f_{qs} , the two-term quasi-symmetry error (Helander & Simakov 2008; Rodríguez *et al.* 2022), and f_{ar}, f_B , which are the residuals from the target aspect ratio (AR) and average magnetic field, respectively. The terms w_{qs}, w_{ar}, w_B are the corresponding weights for each objective. For the results presented here, we used a continuation in Zernike coefficients Rp_{lm} and Zp_{lm} where at each step higher l (and corresponding m) coefficients are allowed to change, and we also included exponential spectral scaling (ESS) for the optimization parameters (Jang *et al.* 2026).

The two-term QS metric is given by,

$$f_{qs} = \frac{(M\iota - N)(\mathbf{B} \times \nabla\psi) \cdot \nabla B - (MG + \iota I)\mathbf{B} \cdot \nabla B}{\langle |\mathbf{B}| \rangle^3} \quad (3.6)$$

where $\mathbf{B}$ is the magnetic field, ∇B is the gradient of the magnitude of the magnetic field, (M, N) are the QS helicity with $(M, N) = (1, 4)$ is chosen for the QH case, ψ is the normalized toroidal flux, I, G are the Boozer toroidal and poloidal currents and $\langle |\mathbf{B}| \rangle$ is the volume average magnetic field magnitude, respectively. It is worth noting that we had to normalize the standard two-term QS metric defined in DESC by the cube of the volume average magnetic field to prevent the optimizer from cheating by reducing the magnetic field strength. This performed much better than tuning the weight for the

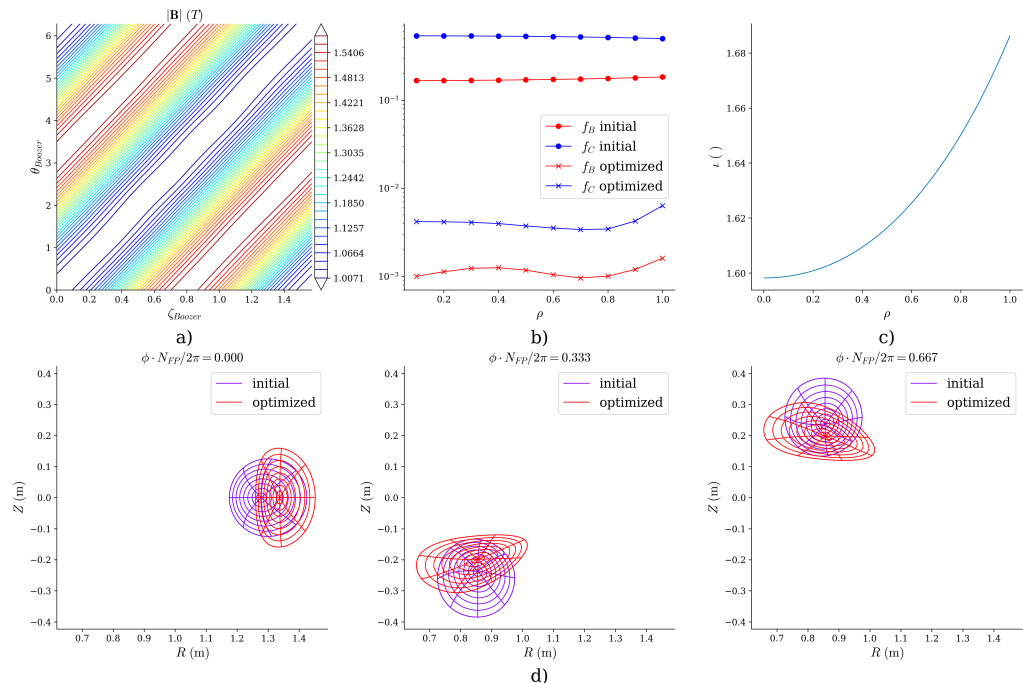


Figure 4: a) $|B|$ level sets in Boozer coordinates for $\rho = 1$ surface, b) Quasisymmetry errors before and after optimization with respect to 2 different metric where f_B is the traditional definition using the Boozer transform and f_c is defined in Equation 3.6 c) Rotational Transform of the optimized equilibria, d) The flux surfaces of the initial and optimized equilibrium.

average magnetic field strength. The importance of the normalization term has also been mentioned in Landreman & Paul (2022), however, in practice, LCFS-based optimizations are less dependent on the inclusion of this term.

Another important distinction from the LCFS optimization is that for the Poincaré variant, we included an additional volume average magnetic field strength objective, which is mostly enforced for the LCFS version by the target aspect ratio, the fixed Rb_{00} mode of the LCFS and the fixed toroidal magnetic flux, Ψ . We have seen that without this penalty, the optimizer sometimes finds a similar solution that has a larger major/minor radius while keeping the aspect ratio the same. Similarly, a volume objective can be added as regularisation for the problem. For this particular example, we observe that the optimizer can find good solutions without needing it explicitly. On a similar point, we didn't fix Rp_{00} since it only constrains the average radius of the Poincaré cross-section without giving any other geometric meaning as Rb_{00} . The inclusion of a fixed Rp_{00} constraint is found to deteriorate the optimization performance significantly. Apart from these differences, the optimization is conducted in the same way with similar initial guess, which has an LCFS of near circular cross sections with axis torsion[†].

The QH optimization results shown in Figure 4 are obtained with the following weights,

[†] More specifically, the initial guess used for the LCFS optimization in Landreman & Paul (2022) is solved by fixed Poincaré boundary condition, and this result is set as the initial guess for the optimization. The extra Poincaré solve didn't change the equilibrium in a distinguishable way.

targets and bounds,

$$\begin{aligned} \langle |\mathbf{B}| \rangle &= 1.26\text{T} & w_B &= 1 \\ AR &= 8 & w_{ar} &= 3 \\ w_{qs} &= 3/|f_{qs0}| \end{aligned} \tag{3.7}$$

where $|f_{qs0}|$ is the initial value of the QS objective without any weights which is 33.3 for this initial guess, and it has helped to reduce the scanned weight combinations. The volume of the initial guess and the final optimized solution are both 0.307 m^3 . The number of field period symmetry is chosen to be $N_{fp} = 4$.

The quasi-symmetry error falls by roughly two orders of magnitude, with $f_{qs} = f_c$ dropping from 0.499 to $6.3\text{E-}3$ at the boundary and by a somewhat larger factor further in, while the aspect ratio and the volume-averaged field strength stay at their targets. The volume-averaged force error normalized by volume-averaged magnetic plasma pressure remains between $1.5\text{E-}5$ and $2.5\text{E-}5$ at every step of the continuation, so the quasi-symmetry is always evaluated on an equilibrium that is well converged in force balance.

4. Conclusion

In this work, the Poincaré cross-section is proposed as an alternative to the LCFS boundary condition for three-dimensional ideal MHD equilibria, and implemented in DESC. Because the condition enters only through the linear constraint matrix $\mathbf{A}$, the rest of the solver and of the optimization machinery is very similar. Prescribing the geometry on a single toroidal plane fixes one linear combination of the toroidal coefficients for each (l, m) and leaves a considerably larger nullspace than a fixed LCFS does, and it is this additional freedom that the solver exploits. We show that solving the equilibrium with the new boundary condition gives lower force errors in a variety of cases. Moreover, the same coefficients can serve directly as design variables, and we used them to obtain a quasi-helical configuration while maintaining force balance to high fidelity throughout the optimization which is crucial for computing the physics objectives reliably.

A few points should be kept in view. The Poincaré cross-section is an algorithmic device rather than a statement about the actual field: it does not guarantee island-free equilibria, and being the less restrictive of the two conditions it makes the local minima of the force-error minimization easier to encounter, which is why a solve started far from the intended configuration can converge to one of very different character. Starting instead from an existing LCFS solution keeps the solver in the intended region, and is the mode in which we find the method most useful. The volume and aspect ratio that a prescribed LCFS carries implicitly are likewise left free, and can be prescribed explicitly as objectives. Residuals should be normalized by a quantity evaluated at the current iterate, since otherwise an expanding equilibrium can lower the reported error by weakening its field rather than by improving force balance; the same consideration applies to the quasisymmetry metric used in the optimization. What these have in common is that they call for objective functions suited to the freedom the boundary condition provides, rather than for the boundary condition to be constrained back towards the LCFS one. Used in this way, the Poincaré boundary condition offers an inexpensive way to reduce the force error that most physics metrics used in stellarator design implicitly assume to vanish.

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6. Data availability statement

The source code to generate the results and plots in this study is openly available on GitHub at <https://github.com/PlasmaControl/DESC> under the `poincare-bc-paper` branch and <https://github.com/PlasmaControl/Poincare-Paper>.

7. Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author Yigit Gunsur Elmacioglu used Anthropic’s Claude Opus 5 in order to increase the readability of the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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